

Tensor for Target Tracking: A Tucker Decomposition Based Method

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Abstract—This paper addresses the continuous-discrete nonlinear target tracking problem, in which the solution to the Fokker-Planck equation (FPE) for state prediction faces the curse of dimensionality. To cope with this problem, we propose a novel tensor decomposition-based method that adopts Tucker decomposition to decompose the high-dimensional PDF of target's state and employs the Fokker-Planck operator to achieve the FPE's solution. Numerical simulations demonstrate the superiority of the proposed method in solving the problem of continuous-discrete nonlinear target tracking.

Index Terms—Fokker-Planck equation, numerical method, nonlinear target tracking, Tucker decomposition.

I. INTRODUCTION

NOWADAYS, nonlinear target tracking plays a crucial role in a wide range of applications [1], including battlefield defense, maritime monitoring, image processing, autonomous drivings [2] and biological systems [3] [4] [5], distributed sensing and information fusion scenarios [6]. The objective of nonlinear target tracking is to recursively estimate the probability density function (PDF) of the state of targets with nonlinear dynamics from noisy measurements [7]. However, the nonlinear target tracking problem becomes increasingly challenging in high-dimensional state spaces due to the so-called curse of dimensionality, in which the computational complexity grows exponentially with the number of state dimensions [1].

Traditional solutions to nonlinear target tracking, such as the Kalman filter and its extended versions, are effective for targets with linear or mildly nonlinear dynamics but are limited by multi-modal and highly nonlinear situation. To overcome the limitation, various methods have been proposed, including

Monte Carlo methods [8] [9], sparse grid discretization methods, tensor decomposition techniques [10] [11]. Monte Carlo methods offer flexibility in handling complex distributions but require a large number of particles to achieve accurate approximations, making them computationally expensive for real-time applications [1]. Sparse grid discretization methods reduce computational complexity but still suffer from exponential growth with increasing dimensions [12] [13].

A promising alternative is the use of tensor decomposition techniques [14], which provides an efficient way to approximate high-dimensional distributions by decomposing them into lower-dimensional components [11]. Among various tensor decomposition methods, Tucker decomposition offers significant advantages in terms of computational efficiency and accuracy [10]. By applying Tucker decomposition to the PDF of target's state, computational burden can be significantly reduced, which poses a great advantage when dealing with nonlinear target tracking problems [15] [16].

Motivated by this, in this paper, we propose a novel nonlinear target tracking method based on Tucker decomposition. Specifically, Tucker decomposition is adopted to decompose the high-dimensional PDF of target's state, then the matrix exponential of the Fokker-Planck operator [17] is employed to solve the Fokker-Planck equation (FPE), which describes the continuous time evolution of the PDF of target state [9]. Subsequently, we achieve the posterior estimation of the PDF, and maintain accurate target tracking.

The remainder of this paper is organized as follows: Section II introduces the formulated problem. Section III presents the proposed method. Section IV provides simulation results, followed by conclusions in Section V.

II. PROBLEM FORMULATION

The state of a target is determined by its dynamic model, for a given target, suppose that its state can be represented by a state vector $\mathbf{x} = [x_1, \dots, x_D]^T \in \mathbb{R}^D$, where D is the

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dimension of the state vector $\mathbf{x}$, and $\mathbb{R}$ denotes real space. The composition of $\mathbf{x}$ can be variables such as the target's speed, acceleration, and position. The time evolution of the dynamics of this target can be described by using the following continuous stochastic dynamic system description [18]:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + \mathbf{G}(\mathbf{x}, t)d\mathbf{w}(t), \quad (1)$$

where $t \in \mathbb{R}$ is the actual time, $\mathbf{f}: \mathbb{R}^D \times \mathbb{R} \mapsto \mathbb{R}^D$ denotes the drift vector, $\mathbf{G}: \mathbb{R}^D \times \mathbb{R} \mapsto \mathbb{R}^{D \times M}$ denotes the matrix of diffusion coefficients and $\mathbf{w}: \mathbb{R} \mapsto \mathbb{R}^M$ is a Brownian motion process with zero mean and covariance $\mathbf{Q}(t)$, and M is the dimension of the Brownian motion process $\mathbf{w}(t)$.

Suppose there exists a sensor such as radar to track this target, and its measurement can be obtained through the following discrete measurements equation with discrete time. Specifically, at time step t_n ,

$$\mathbf{y}_{t_n} = \mathbf{h}(\mathbf{x}(t_n), t_n) + \nu_{t_n}, \quad (2)$$

where $\mathbf{h}(\mathbf{x}(t_n), t_n)$ is used to describe the transfer relationship from the target state to the measurement, which can be a nonlinear function, ν_{t_n} is a white Gaussian noise with covariance $\mathbf{R}$. Equivalently, the measurement process (2) also can be described by a likelihood function $L(\mathbf{y}_{t_n}|\mathbf{x})$.

In order to track the target, the sensor needs to perform iterative nonlinear target tracking process based on measurement equation (2) and dynamic system equation (1). Usually, the target tracking process consists of two steps: the prediction step and the update step.

In the prediction step, suppose that the prior PDF of the target's state $p(\mathbf{x}, t_{n-1}|\mathbf{y}_{t_{n-1}})$, is given at time step t_{n-1} . This PDF then needs to be predicted to the next time step t_n through time evolution, which can be accomplished by solving the following FPE,

$$\frac{\partial p}{\partial t} = - \underbrace{\sum_{i=1}^D \frac{\partial([\mathbf{f}]_i p)}{\partial x_i}}_{\text{drift item}} + \underbrace{\frac{1}{2} \sum_{i,j=1}^D \frac{\partial^2([\mathbf{G}\mathbf{Q}\mathbf{G}^T]_{i,j} p)}{\partial x_i \partial x_j}}_{\text{diffusion item}}. \quad (3)$$

Taking $p(\mathbf{x}, t_{n-1}|\mathbf{y}_{t_{n-1}})$ as the initial condition of the FPE, the new predicted PDF $p(\mathbf{x}, t_n|\mathbf{y}_{t_{n-1}})$ can be obtained by solving the above FPE. In other words, the prediction of the prior PDF can be formulated as the solving of the above FPE, which governs the time evolution of target's PDF under drift and diffusion processes. Specifically, the first term on the right-hand side of FPE (i.e., $\sum_{i=1}^D \frac{\partial([\mathbf{f}]_i)}{\partial x_i}$) represents the deterministic drift effect, while the second term (i.e., $\sum_{i,j=1}^D \frac{\partial^2([\mathbf{G}\mathbf{Q}\mathbf{G}^T]_{i,j})}{\partial x_i \partial x_j}$) corresponds to stochastic diffusion driven by the process noise covariance $\mathbf{Q}(t)$. By solving this partial differential equation, the prior PDF is propagated from t_{n-1} to t_n .

The posterior PDF is dynamically updated through the following Bayesian equation upon receiving new measurements $\mathbf{y}_{t_n}$,

$$p(\mathbf{x}, t_n|\mathbf{y}_{t_n}) = \frac{p(\mathbf{x}, t_n|\mathbf{y}_{t_{n-1}})L(\mathbf{y}_{t_n}|\mathbf{x})}{\int p(\mathbf{x}, t_n|\mathbf{y}_{t_{n-1}})L(\mathbf{y}_{t_n}|\mathbf{x})d\mathbf{x}}. \quad (4)$$

While some finite element methods [11] have been widely adopted to cope with (3), their effectiveness may deteriorate significantly in high-dimensional scenarios. The reason can be attributed to the limitation of high dimension, which induces exponential growth of computational costs, as rigorously analyzed in [11].

To overcome this limitation, we propose a Tucker decomposition-based method. This method eliminates the need for explicit high-dimensional grid construction, enabling efficient computation of the above predicted PDF (3) and posterior PDF (4). Furthermore, the proposed method establishes a computationally tractable framework for solving high-dimensional FPE that conventional discretization methods cannot handle.

III. TUCKER DECOMPOSITION BASED METHOD

A. Brief Introduction To Tucker Decomposition

Tucker decomposition [19] approximates a D -dimensional tensor $\mathcal{T} \in \mathbb{C}^{I_1 \times \dots \times I_D}$ as:

$$\mathcal{T} = \mathcal{A}_{\text{core}} \times_1 \mathbf{A}_1 \times_2 \mathbf{A}_2 \cdots \times_D \mathbf{A}_D, \quad (5)$$

where $\mathcal{A}_{\text{core}} \in \mathbb{C}^{R_1 \times \dots \times R_D}$ is the core tensor, $\mathbf{A}_d \in \mathbb{C}^{I_d \times R_d}$ are factor matrices, and $\times_d$ denotes the mode- d product. To compute these components, we employ the Higher-Order Singular Value Decomposition (HOSVD). For each mode $j \in \{1, \dots, D\}$, the factor matrix $\mathbf{A}_j$ is derived from the R_j leading left singular vectors of the mode- j unfolding $\mathbf{T}_{(j)}$:

$$\mathbf{T}_{(j)} = \mathbf{U}_j \mathbf{\Lambda}_j \mathbf{V}_j^\top \implies \mathbf{A}_j = \mathbf{U}_j(:, 1 : R_j), \quad (6)$$

where R_j is the truncated rank. Finally, the core tensor is calculated as:

$$\mathcal{A}_{\text{core}} = \mathcal{T} \times_1 \mathbf{A}_1^\top \times_2 \mathbf{A}_2^\top \cdots \times_D \mathbf{A}_D^\top. \quad (7)$$

B. FPE Operator

To represent the state uncertainty of a target, we first build a grid that divides each dimension of the state space into small intervals. In this way, the continuous PDF of the target can be discretized into a tensor form. Suppose the state has D dimensions, and the d -th dimension is divided into n_d grid cells. The resulting discretized tensor-form PDF can be written as

$$p(x_{i_1}, \dots, x_{i_D}|\mathbf{y}_{t_{n-1}}) = [p(\mathbf{x}|\mathbf{y}_{t_{n-1}})]_{i_1, \dots, i_D}. \quad (8)$$

This expression means that the value of the tensor-form PDF at grid index $(i_1, \dots, i_D)$ corresponds exactly to the value of the original continuous PDF evaluated at that grid point. In other words, each element in the tensor represents the probability density at one location of the discretized state space.

To mitigate the exponential growth of tensor size in high dimensions—the curse of dimensionality—we employ Tucker decomposition to approximate the high-dimensional tensor via a compact core tensor and factor matrices. This low-rank representation enables efficient computation of the time evolution by solving the FPE (3) using the following operator.

Suppose that the components of both the drift term and the diffusion matrix in (3) can be provided in a separable representation [20]:

$$\begin{aligned} [\mathbf{f}]_i(x_1, \dots, x_D) &= \sum_{k=1}^{K_i} \prod_{d=1}^D f_{d,k}^{(i)}(x_d), \\ [\mathbf{G}\mathbf{Q}\mathbf{G}^T]_{i,j} &= \sum_{k=1}^{K_{i,j}} \prod_{d=1}^D g_{d,k}^{(i,j)}(x_d), \end{aligned} \quad (9)$$

where k is an index denoting the k -th component in the separable representation, respectively. K_i and $K_{i,j}$ indicates the total number of separable components for the drift term and diffusion term along the dimension, both f and g are component functions. According to [20], the drift and diffusion terms are assumed to be time-invariant, then their operators can be further described as:

$$\begin{aligned} -\sum_{i=1}^D \frac{\partial([\mathbf{f}]_i)}{\partial x_i} &= \mathbb{L}^{\text{drift}} = -\sum_{i=1}^D \sum_k^{K_i} \bigotimes_{d=1}^D \mathbb{L}_{d,k}^{(i)}, \\ \frac{1}{2} \sum_{i,j=1}^D \frac{\partial^2([\mathbf{G}\mathbf{Q}\mathbf{G}^T]_{i,j})}{\partial x_i \partial x_j} &= \mathbb{L}^{\text{diffusion}} = \frac{1}{2} \sum_{i,j=1}^D \sum_k^{K_{i,j}} \bigotimes_{d=1}^D \mathbb{L}_{d,k}^{(i,j)}, \end{aligned} \quad (10)$$

where

$$\begin{aligned} \mathbb{L}_{d,k}^{(i)} &= \begin{cases} \text{Diag}(\mathbf{f}_{d,k}^{(i)}) & d \neq i \\ \mathbf{D}_i \text{Diag}(\mathbf{f}_{i,k}^{(i)}) & d = i \end{cases}, \\ \mathbb{L}_{d,k}^{(i,j)} &= \begin{cases} \text{Diag}(\mathbf{g}_{d,k}^{(i,j)}) & d \neq i, j \\ \mathbf{D}_j^2 \text{Diag}(\mathbf{g}_{j,k}^{(j,j)}) & d = j \wedge i = j \\ \mathbf{D}_j \text{Diag}(\mathbf{g}_{j,k}^{(i,j)}) & d = j \wedge i \neq j \\ \mathbf{D}_i \text{Diag}(\mathbf{g}_{i,k}^{(i,j)}) & d = i \wedge i \neq j \end{cases}, \end{aligned} \quad (11)$$

and $\mathbf{D}_i$ or $\mathbf{D}_j$ is the differentiation matrix with respect to the i -th or j -th dimension, $\mathbb{L}^{\text{drift}}$ is the operator for drift terms, and $\mathbb{L}^{\text{diffusion}}$ is the operator for diffusion terms, and

$$[\mathbf{f}_{d,k}^{(i)}]_{n_d} = f_{d,k}^{(i)}(x_{i_d}); \quad [\mathbf{g}_{d,k}^{(i,j)}]_{n_d} = g_{d,k}^{(i,j)}(x_{i_d}). \quad (12)$$

(12) indicates that the discrete point calculation for the component of drift term and diffusion matrix is the same as (8).

Then, the FPE (3) can be simplified as:

$$\frac{\partial \mathbf{p}(t_n)}{\partial t} = \mathbb{L} \mathbf{p}(t_n) = (\mathbb{L}^{\text{drift}} + \mathbb{L}^{\text{diffusion}}) \mathbf{p}(t_n), \quad (13)$$

where $\mathbb{L} = \mathbb{L}^{\text{drift}} + \mathbb{L}^{\text{diffusion}}$. Further, the time-evolution of the tensor-form PDF can be expressed in a matrix-exponential form as:

$$\mathbf{p}(t_{n+1}) = e^{\Delta t \cdot \mathbb{L}} \mathbf{p}(t_n). \quad (14)$$

Based on this, we can perform Tucker decomposition on the PDF on the right side of (14). Then the time-evolution of target's PDF can be written as:

$$\mathbf{p}(t_{n+1}) = e^{\Delta t \cdot \mathbb{L}} (\mathcal{A}_{\text{core}} \times_1 \mathbf{A}_1 \times_2 \mathbf{A}_2 \cdots \times_D \mathbf{A}_D). \quad (15)$$

C. Prediction Step Via Tucker Decomposition

The next step is to update the factor matrix. According to (10), the FPE operator can also be written as a sum of components of multiple dimensions:

$$\mathbb{L} = \sum_{i=1}^D \mathbb{L}_{\text{drift}}^{(i)} + \sum_{i,j=1}^D \mathbb{L}_{\text{diffusion}}^{(i,j)}. \quad (16)$$

Substituting (16) into (15), using LieTrotter splitting to split the matrix exponent into each dimension, and multiplying the split matrix exponent with the factor matrix of the corresponding dimension, yields:

$$\mathbf{A}_{(i)}^{\text{new}} = e^{(\mathbb{L}_{\text{drift}}^{(i)} + \mathbb{L}_{\text{diffusion}}^{(i,\cdot)}) \cdot \Delta t} \mathbf{A}_{(i)}. \quad (17)$$

Thus, the predicted PDF can be described as:

$$\mathbf{p}(t_{n+1}) = \mathcal{A}_{\text{core}} \times_1 \mathbf{A}_{(1)}^{\text{new}} \times_2 \mathbf{A}_{(2)}^{\text{new}} \cdots \times_D \mathbf{A}_{(D)}^{\text{new}}. \quad (18)$$

D. Update Step Via Tucker Decomposition

The likelihood function $L(\mathbf{y}_{t_n}|\mathbf{x})$ can be discretized in the same way. Specifically, we use Tucker decomposition to approximate the likelihood function

$$L(\mathbf{y}_{t_n}|\mathbf{x}) = \mathcal{G}_{\text{core}} \times_1 V_1 \times_2 V_2 \cdots \times_D V_D. \quad (19)$$

Finally, tensor-form PDF is then updated using the Bayesian equation

$$\hat{\mathbf{p}}(t_{n+1}) = \frac{1}{c} \mathbf{p}(t_{n+1}) \odot L(\mathbf{y}_{t_n}|\mathbf{x}), \quad (20)$$

where c is a normalization constant, and $\hat{\mathbf{p}}(t_{n+1})$ is the solution to (4), i.e., $p(\mathbf{x}, t_n|\mathbf{y}_{t_n})$.

For the update of the likelihood at each step, an adaptive energy-based HOSVD truncation is applied. Specifically, for per mode i , only the leading singular components retaining at least $1 - \varepsilon_i$ of the cumulative energy are kept, where ε_i is the threshold. The truncation strategy directly controls both normalization and divergence errors, keeping the low-rank approximation of likelihood stable.

Computation Complexity. For the proposed method, the dominant computation cost of each step arises from modal propagation and the truncated HOSVD re-projection, which together scale as $\mathcal{O}(Dr^3N)$ for a D -dimensional state, grid size N per mode, and per-mode rank r . In contrast, the existing methods such as the Kronecker-based method [20] has exponential cost $\mathcal{O}(N^D)$ in both time and memory, and the TT-based method [21] scales as $\mathcal{O}(DNr_{\text{TT}}^3)$, where r_{TT} is the truncated rank of the TT decomposition.

IV. PERFORMANCE EVALUATION

In this section, simulation experiments are conducted to verify the effectiveness of the proposed method.

We consider a target moving in a two-dimensional plane with the four-dimensional state vector

$$\mathbf{s} = [x, y, v_x, v_y]^T,$$

where (x, y) denotes the position and (v_x, v_y) denotes the velocity in Cartesian coordinates. we consider the coordinated-turn (CT) scenario with a known turn rate ω . The continuous-time stochastic dynamics are described as:

$$\text{CT: } ds(t) = \underbrace{\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\omega \\ 0 & 0 & \omega & 0 \end{bmatrix}}_{\mathbf{F}_{\text{ct}}(\omega)} s(t) dt + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}_{\mathbf{G}_{\text{ct}}} \sigma_{\text{ct}} d\mathbf{w}(t), \quad (21)$$

where $\mathbf{w}(t) = [w_x(t), w_y(t)]^T$ is a standard two-dimensional Brownian motion process, the process noise levels $\sigma_{\text{ct}} = 0.05$ with a fixed turn rate $\omega = 0.1$ rad/s. The total simulation time is $T = 100\text{s}$ with a time interval $\Delta t = 0.1$ s.

The initial state follows a Gaussian distribution $p_0(\mathbf{s}) = \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, where $\boldsymbol{\mu}_0 = [2, 2, 0.8, 0.3]^T$ and $\boldsymbol{\Sigma}_0 = \text{diag}(4, 4, 0.5, 0.5)$. The 4-D state space is discretized uniformly with $x, y \in [-30, 30]$ and $v_x, v_y \in [-3, 3]$, using $N = 50$ nodes per dimension. Observations are generated as:

$$\mathbf{z}_k = \begin{bmatrix} x_k \\ y_k \end{bmatrix} + \boldsymbol{\nu}_k, \quad \boldsymbol{\nu}_k \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{obs}}^2 \mathbf{I}_2), \quad \sigma_{\text{obs}} = 1.0. \quad (22)$$

At each update, HOSVD with an energy keep of 99.5% adaptively truncates per-mode ranks, effectively filtering numerical artifacts while retaining the dominant mass transport.

To demonstrate the advantages of the proposed method, we compare the proposed method with two existing methods, namely: TensorTrain (TT)-based method [21] and Kronecker-based method [20]. We use RMSE and wall-clock time as the criteria to evaluate the performance of these methods.

The results after Monte Carlo trails are shown in Fig. 1, Fig. 2 and Table I. Fig. 1 depicts a typical CT trajectory ($\omega = 0.1$ rad/s) together with the noisy observations and the estimated trajectory, Fig. 2 shows the RMSE results of these methods in the same CT scenario, and Table I shows average per-step wall-clock runtime and average RMSE comparison of these methods.

TABLE I
AVERAGE WALL-CLOCK RUNTIME AND AVERAGE RMSE OF DIFFERENT METHODS.

Method	Runtime (s/step)	Average RMSE
Proposed Method	1.1	0.718
TensorTrain-based Method	0.7	0.739
Kronecker-based Method	0.9	7.345

Fig. 1 shows that the proposed method accurately tracks the target trajectory despite noisy observations. As depicted in Fig. 2, the proposed method achieves the lowest and most stable RMSE, avoiding the numerical instabilities (jumps) observed in the TT-based (19s, 26s) and Kronecker-based (64s) methods. Table I compares the computational costs. Although the proposed method has lower theoretical complexity, its wall-clock time is slightly higher due to constant factors in

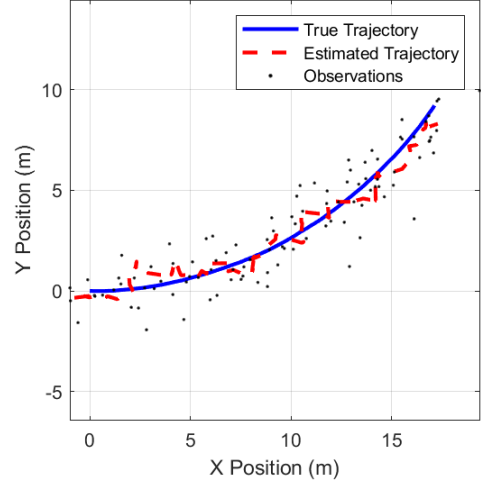


Fig. 1. True/estimated trajectories and observations in the CT scenario.

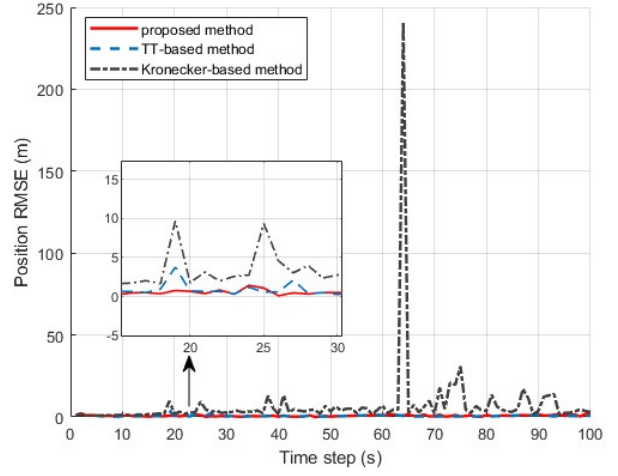


Fig. 2. Position RMSE over time in the CT scenario.

repeated HOSVD truncations and intensive mode-wise matrix operations during the Tucker update.

Overall, although the proposed method takes slightly longer in runtime, its performance is superior to the existing algorithms in terms of both estimation accuracy and estimation stability.

V. CONCLUSION

This paper proposes a novel Tucker decomposition-based nonlinear target tracking method for efficient computation of high-dimensional state PDF. The proposed method overcomes computational challenges in high-dimensional tracking while maintaining estimation accuracy. Future work will explore tensor-based tracking for maneuvering and multiple targets, to achieve fast convergence, and high accuracy of estimated targets' trajectories.

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